Introduction to Computational Physics (CS201)

December 4, 2025

1 Course Placement

Introductory Computational Physics is a core course for students taking a B.Tech (Honours) degree in ICT with a minor in Computational Science.

2 Course Format

The course will consist of 3 hours of lectures and 3 hours of laboratory work each week.

3 Course Content

This course is intended to teach students how to formulate and solve a large number of physical problems mathematically. A large number of analytical and numerical methods will be covered in this course. The areas of physics from which the problems will be selected will be classical mechanics, nuclear processes, oscillations and waves, central forces and planetary motion and basic quantum mechancis. The emphasis will be on formulating the problem mathematically and then using appropriate analytical and numerical methods to arrive at the solution. This is mainly meant to be a course on using calculus to solve physical problems. Consequently, a large part of the course will be focused on differential equations and integrals.

4 Textbooks and References

Textbook
Computational Physics, N. J. Giordano, Prentice Hall, New Jersey

Selected Reference Books

1. *An Introduction to Computational Physics*, Tao Pang, Cambridge University Press,
2. *Modern Physics*, Kenneth S. Krane, Wiley,
3. *Elementary Numerical Analysis*, K. Atkinson and W. Han, Wiley,
4. *Mathematical Methods for Physicists*, G. B. Arfken and H. J. Weber, Harcourt, New Delhi.

Other reference books will be suggested in class from time to time, as appropriate.

5 Assessment

All scheduled in-semester examinations and the end-semester examination will count towards 70% (15% + 20% + 35%) of the total credit, with the remainder (30%) of the credit being from the laboratory component.

6 Course Outcomes

Students will learn to formulate a physical problem and solve it, using analytical and computational methods. Problem-solving will be the main focus of this course. After completion, the students should be able to,

- To understand and analyse the physical world, using a basic set of principles.
- To design, model and investigate complex engineering problems.

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
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7 Course material and Tentative Lecture Schedule

7.1 A Survey of Physical Concepts

Time Allotted (5 Lectures)

- Length scales, fundamental constants, macroscopic and microscopic physics.
- Classical Physics: Newton's Laws and Maxwell's equations.
- Quantum Physics: Basic principles.

7.2 Mathematical Methods

Time Allotted (15 Lectures)

- Differential equations and their uses in modeling physical systems.
- Analytical exact and approximate methods of solution.
- Discretisation of derivatives.
- Euler and Runge-Kutta methods of solving discretised equations.
- Partial differential Equations
- Vector Calculus

7.3 Physical Systems

Time Allotted (20 Lectures)

- Radioactivity and nuclear processes
- Classical Mechanics: Particle motion, simple dynamical systems, dissipative systems
- Oscillatory motion, Damped oscillations
- Central Force
- Gravitation and planetary motion
- Two-body and three-body problem.
- Autonomous systems of ordinary differential Equations
- Wave equation and Diffusion equation
- Schroedinger equation, basic quantum mechanics

8 Objectives and Learning Outcomes

The objectives of this course is to teach the students how to come up with a mathematical formulation of a physical problem and then select appropriate methods to solve it. Laboratory sessions will mainly be used to supplement the lectures, giving students an opportunity to put theory into practice.

9 Course Credits

4.5 $(3 - 0 - 3 - 4.5)$

10 Evaluation and Attendance

Attendance during laboratory sessions is compulsory. Students are expected to submit assignments which will comprise of work done during the laboratory session, along with analytical work where applicable.

11 Laboratory Work

Lab sessions are compulsory and a record of attendance will be kept. Lab work will contribute 30% of the total grade.